Simulation study: mixture of experts model

The Tien Mai & Zhi Zhao

Last updated: 24 November, 2025

Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 3$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.35, 0.40, 0.25)$
- Degrees of freedom $\boldsymbol{\nu} = (8, 12, 3)$
- Scale matrices of the Wishart distribution Σ_1, Σ_3, Σ_3
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2) + \pi_3 \text{Wishart}(\nu_3, \Sigma_3)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(mewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.35, 0.40, 0.25),
  nus = c(8, 12, 3),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 3

# run a mewishart model
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
fit <- mewishart(S_list,
  K = my_K_intial,
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
end_time <- Sys.time()
elapsed_time <- end_time - start_time
```

```
# posterior summaries
colMeans(fit$pi)
```

```
[1] 0.45895532 0.04315591 0.49788877
```

```r
apply(fit$nu, 2, mean)
```

```
[1] 10.830191 11.146458  4.429139
```

```r
# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean <- Reduce("+", fit$Sigma) / length(fit$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean[, , 1])
```

```
          [,1]      [,2]
[1,] 2.1413601 0.4694115
[2,] 0.4694115 1.5746822
```

```r
print(sigma_posterior_mean[, , 2])
```

```
           [,1]       [,2]
[1,]  1.0046334 -0.8349913
[2,] -0.8349913  1.2083814
```

```r
print(sigma_posterior_mean[, , 3])
```

```
          [,1]      [,2]
[1,] 1.2138781 0.4253603
[2,] 0.4253603 1.3718571
```

```r
as.data.frame(Sigma_list)
```

```
   X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2  2.0  0.6  4.0  0.2
2 0.2 0.7  0.6  1.5  0.2  3.0
```

```r
(a1 <- apply(sigma_posterior_mean, MARGIN = 3, function(aa) min(sapply(Sigma_list, function(x) sum((aa 
```

```
[1] 0.05966684 2.65552059 0.69540631
```